

>> muscle connections

Proper positioning of the arms, legs, and head helps us to physically find the link between muscle and connective tissue. Using focus and intent when we line these extremities up with the torso gives us a powerful tool for changing body posture and developing flexibility.

The science of biomechanics identifies various structural body connections and physical forces that are involved in body function. In order to devise appropriate exercises, it is necessary to use our knowledge of the nature of our body parts (how plastic, or changeable, the various components are) to create the effect we need. Three important structural connections in the body that we have to consider are the “X” model, the inner unit, and the lateral system.

The “X” model

The “X” model shows the connection between what is going on externally and the inner unit (see below). It shows how the limbs are connected with each other and how these connections pass right through the inner unit. Think deep; think three-dimensional. The right arm, for example, is connected to the left leg and vice versa. The positioning of the head, which can weigh up to 15lb (6.8kg), is also important. Tipping it in any direction activates an intricate system of overlapping muscles that both bind the head into the trunk and yet allow a marvellous telescoping range to the neck.

The inner unit

Various groups of muscles form the inner unit. These are the muscles at the bottom of the torso (the pelvic floor), the deep abdominal muscles, the transverse abdominals at the sides of the abdomen, the deep low back muscles, the multifidi (a group of

>> pulling it all together

- **Coordination** between opposing limbs and the trunk is demonstrated by the “X” model concept.
- **Precision** in stretching is created by achieving stabilization of the inner unit, which provides a firm foundation.
- **Elongation of the lateral system** promotes symmetry and balance.

muscles either side of the spine), and the muscles deep inside the rib cage (the diaphragm).

Working the muscles of the inner unit correctly—with good form—promotes low back and pelvic health. The exercise instructions also help you to use the inner unit as a stabilizing foundation, giving more precision when you stretch the external parts.

The lateral system

The lateral system connects the muscles and fascia (see p.12) that run down the sides of the body. Think of it as a long road running from the triceps in the upper arm, past the armpit, down the side of the ribs and waist, extending down the side of the leg past the thigh and shin, and ending at the side of the foot. This lateral system is often overlooked, but opening it through stretching is key to balancing the body and improving posture.